Statistical Analysis Report: Income and Sex in the 1994 U.S. Census

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Dataset: UCI Adult (Census Income) training extract (32,561 rows, 15 columns)

Project: Conduct a Statistical Analysis Using Python capstone (Adult income analysis)

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Overview

I analyzed the UCI Adult (Census Income) dataset (Becker & Kohavi, 1996) to test whether sex is associated with earning over $50K. The dataset is publicly available at https://archive.ics.uci.edu/dataset/2/adult. Of its 32,561 records, 7,841 (24.08%) sit over the $50K line.

This report covers the descriptive statistics, four visual models, and one chi-square test of independence. The result of this analysis points to a clear discrepancy between men and women: men in this data earn over $50K far more often than women, 30.57% against 10.95%. The chi-square test attributes the gap to far more than sampling noise, with a p-value of about 1e-332. The effect size, Cramér's V, is 0.2160, a small-to-moderate association.

Dataset Description

As noted in the Overview, the analysis uses the UCI Adult training extract (Becker & Kohavi, 1996). It is a real public extract of the 1994 U.S. Census, hosted by the UCI Machine Learning Repository. Each row is one person, and income is the binary target of whether that person earns over $50K per year.

The table has 15 columns and 32,561 rows: 6 are numeric (age, fnlwgt, education-num, capital-gain, capital-loss, and hours-per-week) and 9 are categorical. The test rests on the categorical pair sex and income, and the descriptive work follows the two numeric variables most relevant to the outcome, age and hours-per-week.

Three columns carry missing values (workclass 1,836, occupation 1,843, and native-country 583, 4,262 cells in total). I keep the missing values in place, because data cleanup is not the purpose of this project. The sex and income columns are complete: all 32,561 rows carry a recorded value for both. The test below therefore uses every row, and the missing cells in the other columns play no part in it.

The load check also flags 24 exact duplicate rows. I keep them, because a census extract

can legitimately hold two people with identical recorded attributes. One caution applies further down: the source is a household survey, and the relationship field lists husband, wife, and own-child entries, so the rows may not be independent.

Methods

Lusa et al. (2024) set out the initial-data-analysis steps that should precede inference: data cleaning, a review of the missing values, and checks on the distribution and shape of each variable. The argument is written for longitudinal studies, but the steps apply to any dataset. Cleanup stays out of scope, which the Dataset Description records, so the screening before any test reduces to two steps: the load check covers the column types, the missing values, and the duplicates, and the descriptive statistics run the shape checks on age and hours-per-week. That screening is the full data preparation, and its results shaped the choices below: a right-skewed age and two unbalanced variables (income and sex).

For the question itself I use a chi-square test of independence. The question asks whether sex and income are associated rather than how much one predicts the other. Both variables are categorical, which is exactly what the test expects. The p-values come from a standard approximation that is only reliable when the expected counts are large (Agresti, 2007), and the expected counts here (checked in the Results) sit far above that level.

The null hypothesis (H0) is that sex and income are independent. The alternative (H1) is that they are associated. I fixed the significance level at alpha = 0.05 before the test.

A chi-square test is non-directional: it can say whether an association exists, not which way it runs, so I read the direction of the difference from the observed counts and rates rather than expecting the test to supply them. A second assumption is that every row is a separate observation. That one is not fully true here: the source is a household survey, records can share a household, and the p-value therefore reads only as approximate (Agresti, 2007). The Limitations section takes up that concern.

SciPy's chi2_contingency applies the Yates correction by default to a two-by-two table. I

report the uncorrected Pearson statistic (the effect size builds on it) and note the corrected value in the Results. As an effect size I report Cramér's V, a 0-to-1 measure of the strength of an association (Ben-Shachar et al., 2023). The p-value alone does not show that strength and is easy to misread as the probability that the null hypothesis is true, so the p-value and the effect size travel together (Greenland et al., 2016).

All computations used Python 3.12.14 with pandas 3.0.6, NumPy 2.5.3, and SciPy 1.18.1 (Virtanen et al., 2020), and the figures used Matplotlib 3.11.2 and Seaborn 0.13.2. The notebook runs top to bottom from a clean kernel, writes the four figures itself, and prints every number the report quotes. requirements.txt pins the exact package versions, and the notebook's Reproducibility section documents the whole rebuild with uv step by step, from the environment creation through the executed notebook.

Results

The descriptive statistics show the shape and balance of the four relevant variables:

- age is right-skewed: the mean is 38.5816, the median is 37.0, the standard deviation is 13.6404, the interquartile range is 28.0 to 48.0, and the skewness is 0.5587.
- hours-per-week concentrates at a 40-hour week: the mean is 40.4375, the median is 40.0, the standard deviation is 12.3474, the interquartile range is 40.0 to 45.0, and the skewness is 0.2276.
- income is unbalanced, with 7,841 people (24.08%) over $50K.
- sex is uneven, with 21,790 men and 10,771 women.

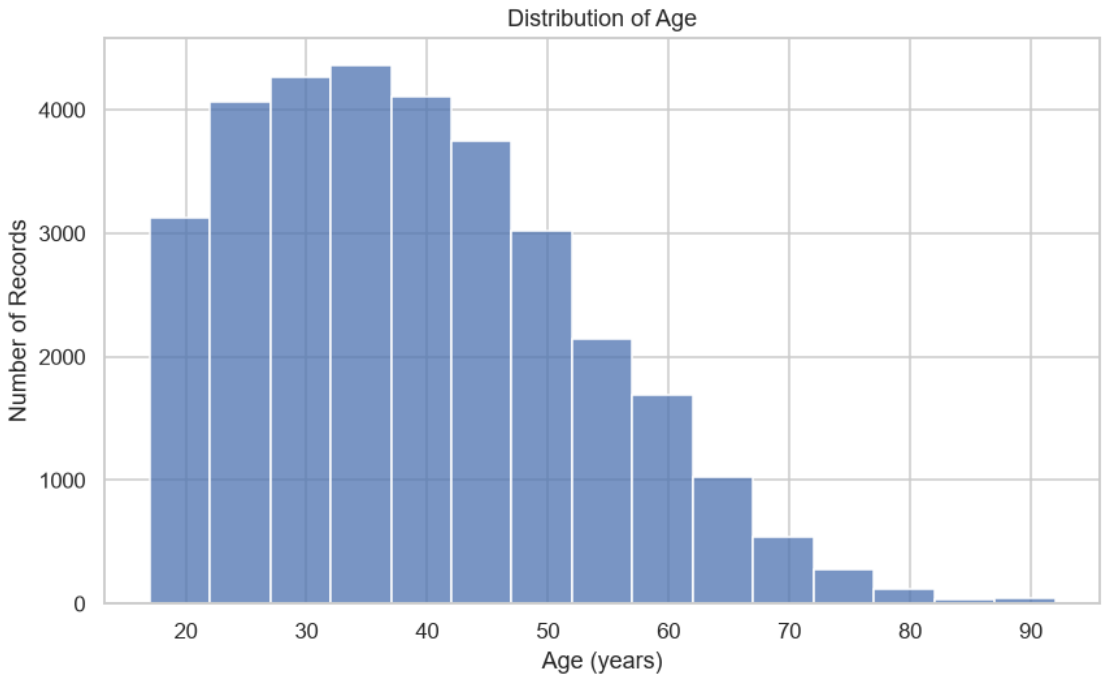


Figure 1. Distribution of age, showing the right skew.

The curve in Figure 1 falls the way the skewness number predicts, but the last populated bin needs its own reading. The notebook's supporting counts show it holds 47 records, with 43 at the 90 cap, 3 at 88, and 1 at 87. The data appear top-coded at 90 (respondents beyond the cap recorded as 90), so that bump reflects the cap rather than a natural peak of older respondents.

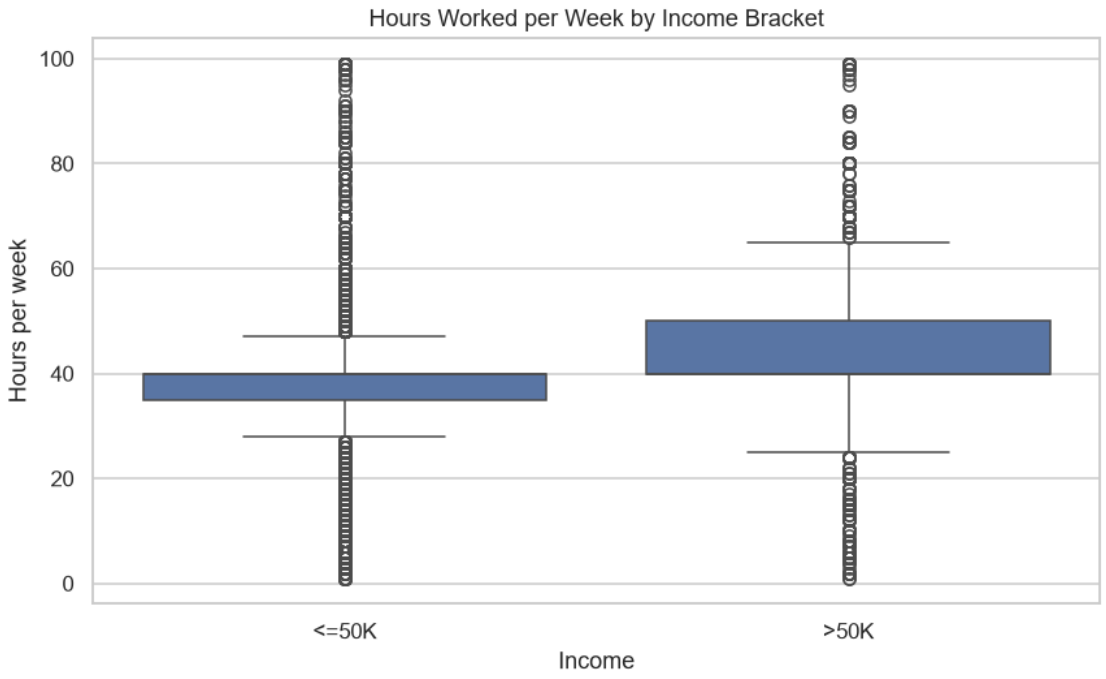


Figure 2. Hours worked per week by income bracket.

Figure 2 compares how weekly hours are spread between the two income groups.

- **Median:** 40 hours in both groups.
- **Middle range:** 40 to 50 hours in the over-$50K group, compared with 35 to 40 hours in the at-or-below-$50K group.
- **Past 40 hours:** 49.2% in the over-$50K group, compared with 23.2% in the at-or-below-$50K group.

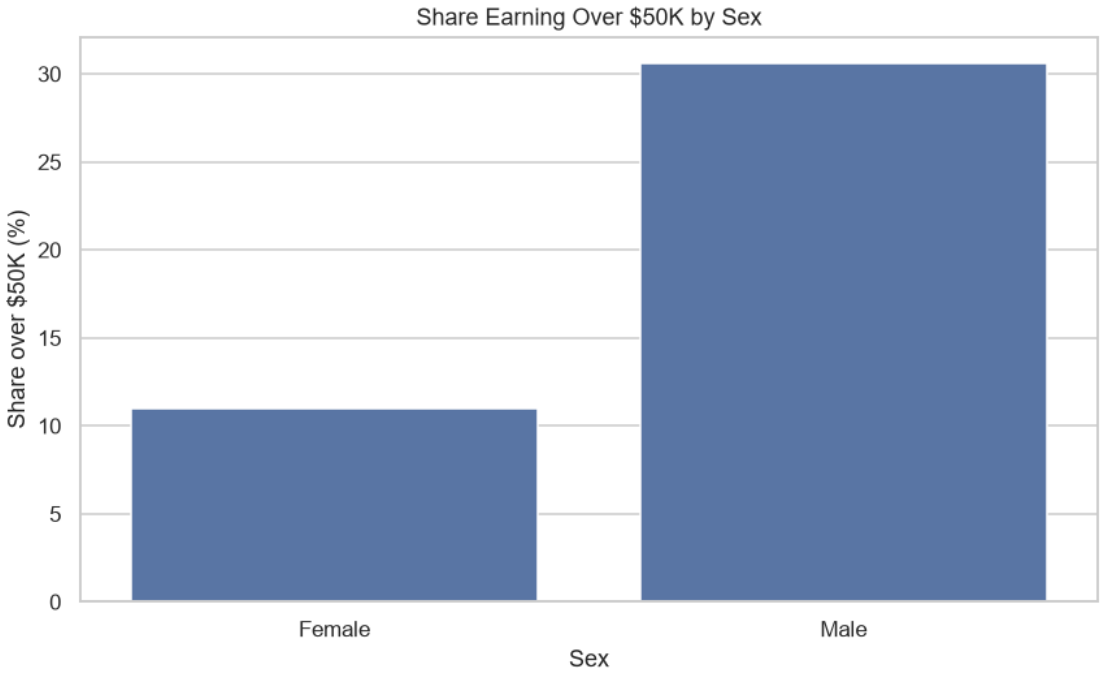


Figure 3. Share earning over $50K by sex.

Figure 3 shows the income rate by sex, the clearest single view of the question. 30.57% of men earn over $50K against 10.95% of women, a gap of 19.63 percentage points.

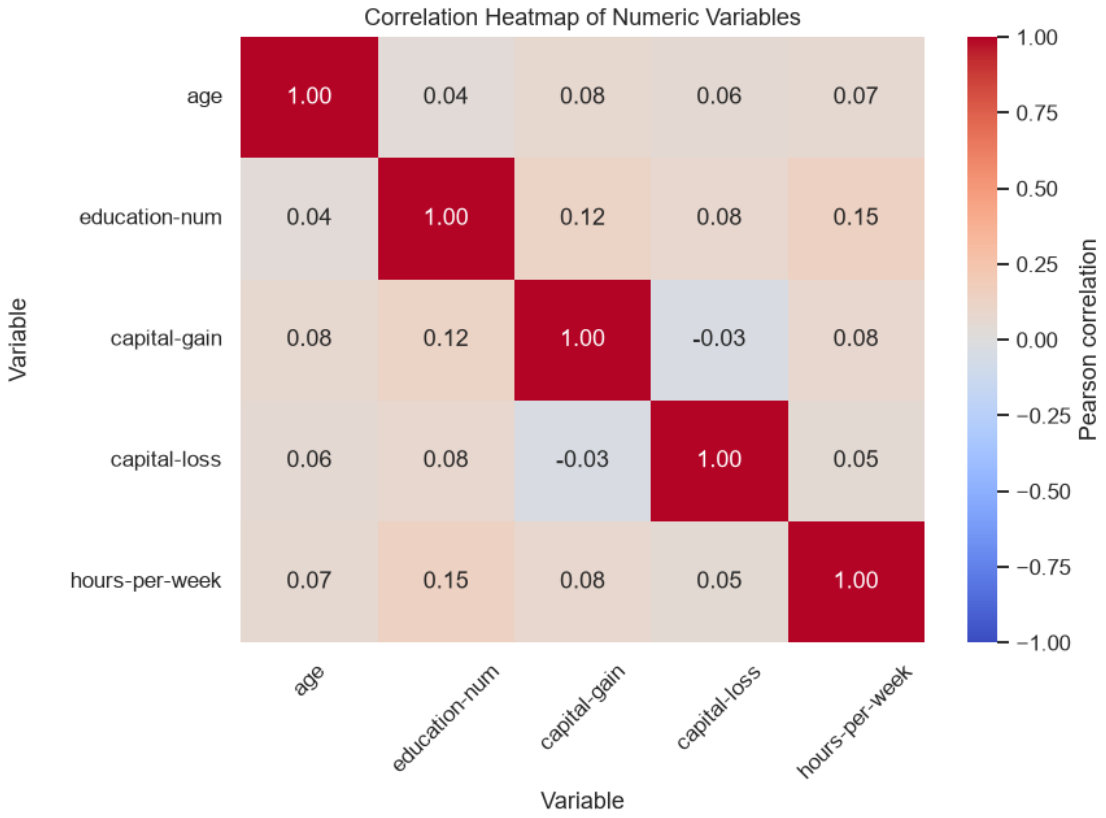


Figure 4. Correlation heatmap of the numeric variables.

Figure 4 shows how the plotted numeric variables move together.

- education-num and hours-per-week are the strongest link: 0.1481.
- income is categorical and outside the matrix, so this view does not directly answer the income question.
- fnlwgt is omitted because it is a census sampling weight, not a measured attribute.
- capital-gain and capital-loss are mostly zeros (91.67% and 95.33%), so their correlations are near-degenerate.

The chi-square test rejects independence.

- Pearson statistic: 1518.8868; Yates-corrected: 1517.8134; degrees of freedom: 1.
- p-value: about 1e-332, far below 1e-300.

- Smallest expected count: 2593.76, so the approximation is safe.
- Effect size: Cramér's V 0.2160; V squared 0.0466 (4.7%).

The contingency table is:

Sex	At or below $50K	Over $50K
Female	`9,592`	`1,179`
Male	`15,128`	`6,662`

The over-$50K rate is 19.63 percentage points higher for men than for women. Its 95% confidence interval spans [18.78, 20.48] (Agresti, 2007). The interval and the p-value both assume independent observations, so possible household clustering makes both approximate.

Interpretation for a Non-Technical Audience

In this dataset, men were much more likely than women to earn over $50K. About three in ten men were above the threshold, against about one in ten women, a gap of about 19.63 percentage points.

A gap of 19.63 points in a sample of 32,561 would almost never appear by chance if sex and income were unrelated, and the chi-square test confirms the gap (p about 1e-332). The effect size adds the scale: 0.2160 is a small-to-moderate value on the 0-to-1 scale, and sex alone explains only about 4.7% of the variation in who is above the line. Sex is one of many differences between the two groups, not the whole of the difference.

The result is an association, not proof that sex caused the difference in earnings, and it is tied to 1994 and to the United States. The Limitations section lists the other differences between the groups and the other caveats.

Limitations and Potential Bias

This is an observational comparison, so the result is an association, not proof of sex causing the difference.

The 1994 U.S. Census extract and its undocumented sampling design limit how far the

result generalizes. The $50K threshold is nominal and not inflation-adjusted, and sex is recorded as a binary field.

Missingness in three columns and the possibility of household-linked rows make the observed rates, p-value, and confidence interval approximate. A follow-up model using education-num, occupation, and hours-per-week, while accounting for those issues, would be a useful next step.

References

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